1145 ▸ 1245

Modern copy of *Tabula Rogeriana*, al-Idrisi's ancient world map

1154

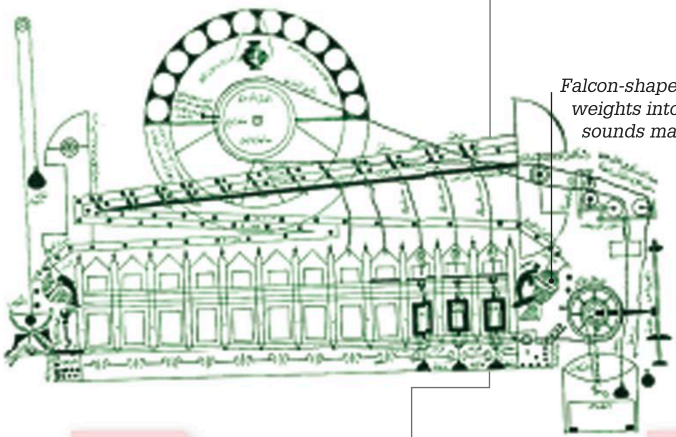
Striking clocks

Arab engineer Muhammad al-Sa'ati constructed the first striking clock in Damascus, Syria. Like many early clocks, it was powered by water. In 1203, his son Ridwan gave a detailed description of the clock's mechanism.

1154

Al-Idrisi's world map

Arab scholar Muhammad al-Idrisi was commissioned by King Roger II of Sicily to compile a world map. It took 15 years to complete and was inscribed on a 6.5-ft- (2-m-) wide silver disc. The most accurate map of its time, it was accompanied by a book detailing all the lands it portrayed.



Falcon-shaped figures released weights into a metal vessel, the sounds marking the hours.

Water power turned the ropes and pulleys that set al-Sa'ati's clock in motion.

1180

Vertical windmills

The first windmills with sails mounted on a vertical tower were introduced in Europe. The spinning of the sails caused a shaft to rotate, which operated hammers used to grind grain. Earlier windmills, developed in Persia (now Iran), had been horizontal, with rectangular sails rotating around an upright shaft.

1145 1165 1185

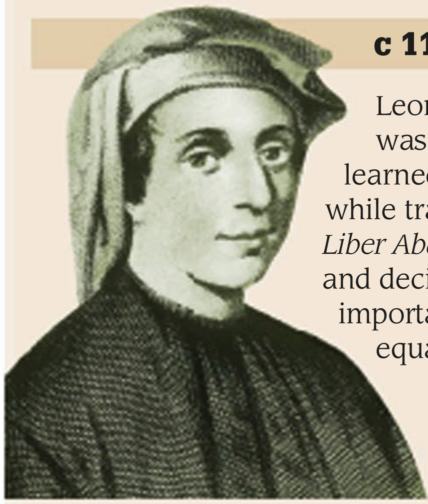
1160

The first printed map

The earliest printed map in the world, the *Shiwu Guofeng dili zhi tu* (Geographic Map of Fifteen States), appeared around this date. It was printed from woodblocks and showed parts of western China. It was published in the *Liu jing tu*, a Chinese encyclopedia. The map listed place names, including rivers and 15 provinces.

In 1150, Indian mathematician Bhaskara II proved that numbers have two square roots, one positive and one negative.

c 1170–1250 FIBONACCI



Leonardo Bonacci, nicknamed Fibonacci, was a merchant from Pisa, Italy, who learned much about Arabic mathematics while trading in North Africa. His book *Liber Abaci* introduced Arabic numerals and decimal notation to Europe. He also did important work in solving certain algebraic equations and in number sequences.

Fibonacci's sequence

Fibonacci described a sequence, later named after him, in which each number is the sum of the two numbers before it (0, 1, 1, 2, 3, 5, 8, 13, and so on). Scientists found mapping a series of squares whose area corresponds to the numbers, and then connecting them, draws a spiral shape often seen in nature—such as a snail's shell.

The number sequence can be shown as a series of boxes.

Connecting the opposite box corners draws a spiral shape.